

Lab 4: VibeCoding — Connect Four vs AI

Software Engineering Laboratory Submission — Task 1 LLM Chat History

STUDENT REPOSITORY	TARGET FOLDER	AI CODING ASSISTANT	MILESTONE
abhiraamadiga/SE_submissions	Lab-4	Antigravity IDE (Gemini)	Task 1 — Complete Win Detection

Task 1 Overview & Deliverables

Task	Requirement	File Modified	Status
Task 1	Complete Win Detection (Diagonal Wins)	board.py	Completed & Verified

Objective: Make win detection recognise every four-in-a-row direction, including both diagonals, without breaking horizontal and vertical wins. Shorter sequences (< 4) do not count.

Prompt and Response History

User Prompt 1

"C:\Users\abhi1\Downloads\Lab_4_VibeCoding_Student_handout (1).pdf"

this contains instructions i want u to rename 15_connect_four to Lab-4 and also wait until i tell u to make changes and add this repo as remote git remote add origin https://github.com/abhiraamadiga/SE_submissions.git

dont do any changes in the 15_connect_four
dont push or add anything

AI Assistant Response

I have completed the initial setup as requested:

1. **Renamed directory:** 15_connect_four has been renamed to Lab-4 .

2. **Configured remote:** The remote origin is set to https://github.com/abhiraamadiga/SE_submissions.git .

3. **No changes made to code:** The files inside Lab-4 remain completely untouched.

4. **No git add / git push:** Nothing has been staged or pushed.

Per the lab handout, please record your 10-second video of the game running before changes (demonstrating the broken diagonal win detection defect). Standing by for your instructions!

User Prompt 2

i made 10-seconds video whts the nexr step?

AI Assistant Response

Great job! Now that you have recorded the "before" video, here are the next steps according to the lab instructions:

1. **Save your "before" video:** Place your recorded 10-second video into `Lab-4/videos/before.mp4`.
2. **Task 1 — Complete Win Detection (`board.py`):**
 - Currently, `winner()` only checks horizontal `(0, 1)` and vertical `(1, 0)`.
 - Add both diagonal directions: `(1, 1)` (down-right) and `(1, -1)` (down-left).
3. Followed by Tasks 2–4 (Game termination, AI decision-making, and move-level feedback).

User Prompt 3

Complete task 1

AI Assistant Response

I have implemented and verified **Task 1 — Complete win detection** in `board.py` !

1. Defect Analysis

In the starter code of `board.py` , `winner(token)` only defined vectors for horizontal and vertical directions:

```
- directions = [(0, 1), (1, 0)] # Missing diagonals!
```

As a result, any four-in-a-row formed diagonally was ignored, allowing the game to continue incorrectly.

2. Code Modifications (`board.py`)

Updated `Board.winner()` to check all four directions from each cell:

```
+ directions = [  
+     (0, 1), # horizontal  
+     (1, 0), # vertical  
+     (1, 1), # diagonal down-right (\)  
+     (1, -1), # diagonal down-left (/)  
+ ]  
for r in range(ROWS):  
    for c in range(COLS):  
        if self.grid[r][c] != token:  
            continue  
        for dr, dc in directions:  
            if all(  
                0 <= r + i * dr < ROWS and  
                0 <= c + i * dc < COLS and  
                self.grid[r + i * dr][c + i * dc] == token  
                for i in range(4)  
            ):  
                return True  
return False
```

3. Verification & Unit Tests

Tested all conditions against the rubric:

- ✓ **Horizontal 4-in-a-row:** Detected as win.
- ✓ **Vertical 4-in-a-row:** Detected as win.
- ✓ **Diagonal down-right (\):** Detected as win.
- ✓ **Diagonal down-left (/):** Detected as win.
- ✓ **3-in-a-row:** Correctly ignored (returns `False`).

Done when criteria satisfied: All four directions are detected and shorter sequences do not count. Task 1 is ready to commit!

User Prompt 4

what was the bug ?

AI Assistant Response

The primary bug in the starter code was in `board.py` : **Missing Diagonal Win Detection.**

- **The Bug:** The original `winner()` method had:

```
directions = [(0, 1), (1, 0)] # Only horizontal and vertical
```

- **The Impact:** When either player or AI placed four discs diagonally (`\` or `/`), the board failed to recognize the win, causing the game to continue indefinitely.
- **The Fix:** We added `(1, 1)` for down-right and `(1, -1)` for down-left diagonal vectors, ensuring all 4 directions are checked without breaking existing horizontal/vertical win checks.